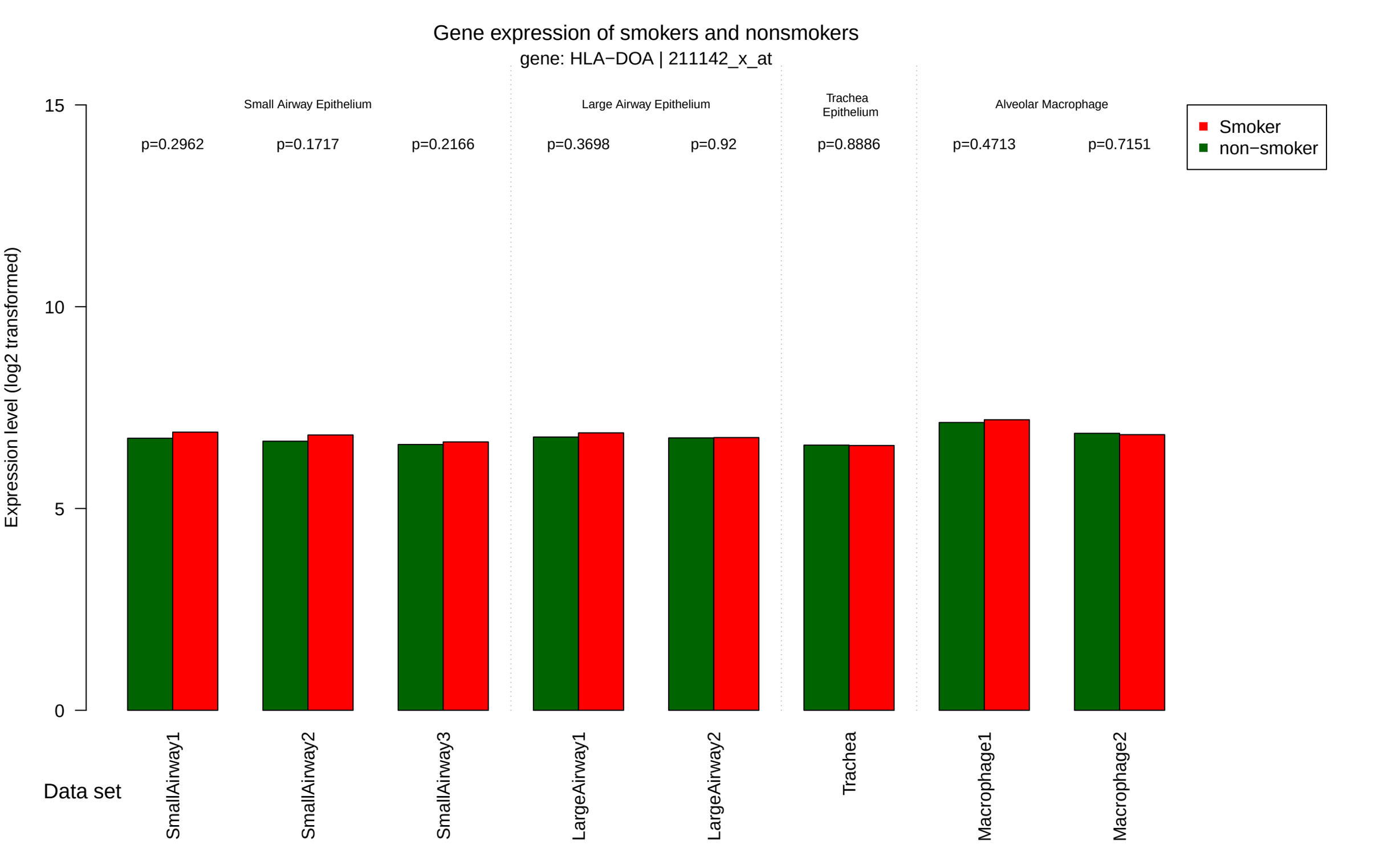
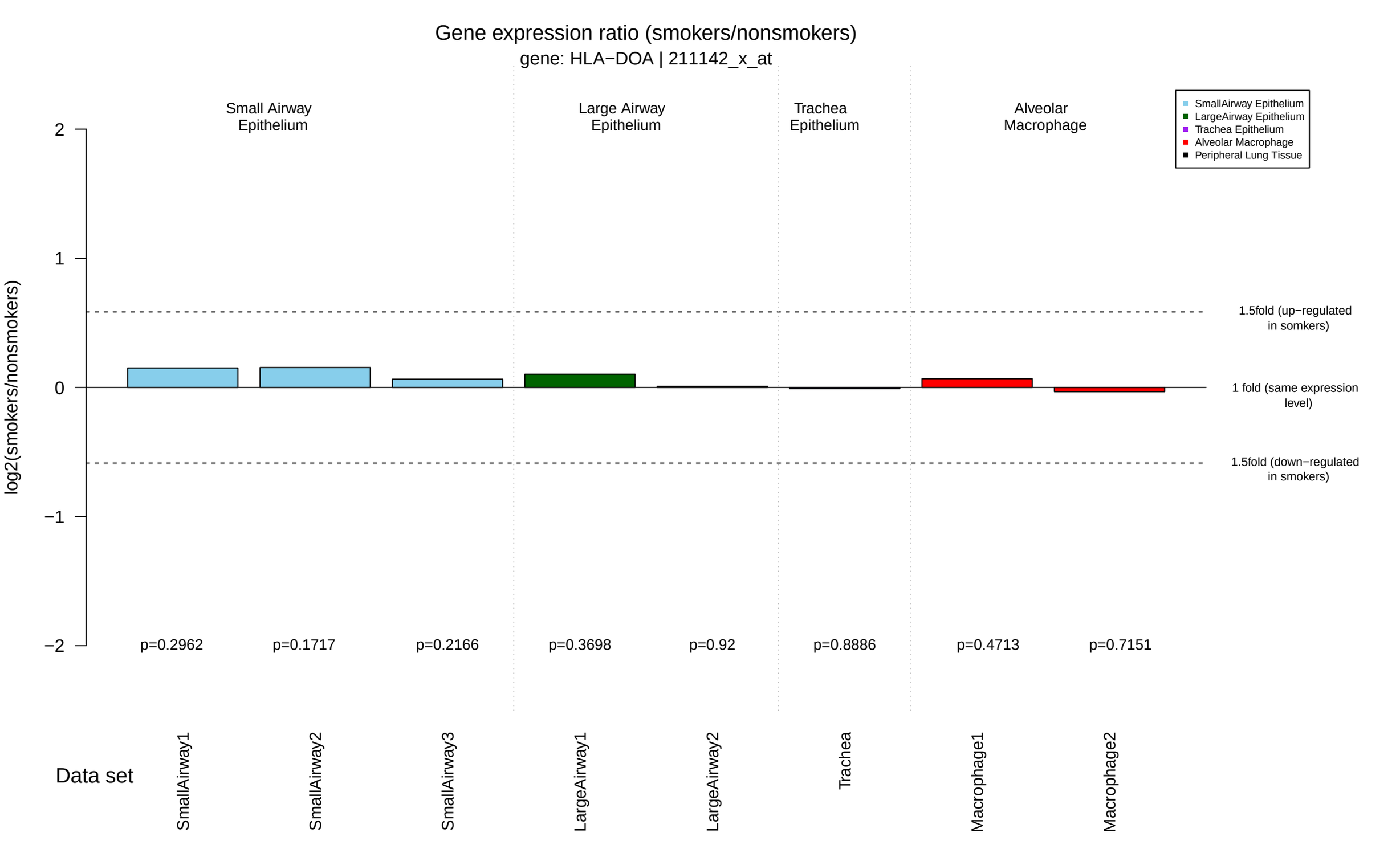
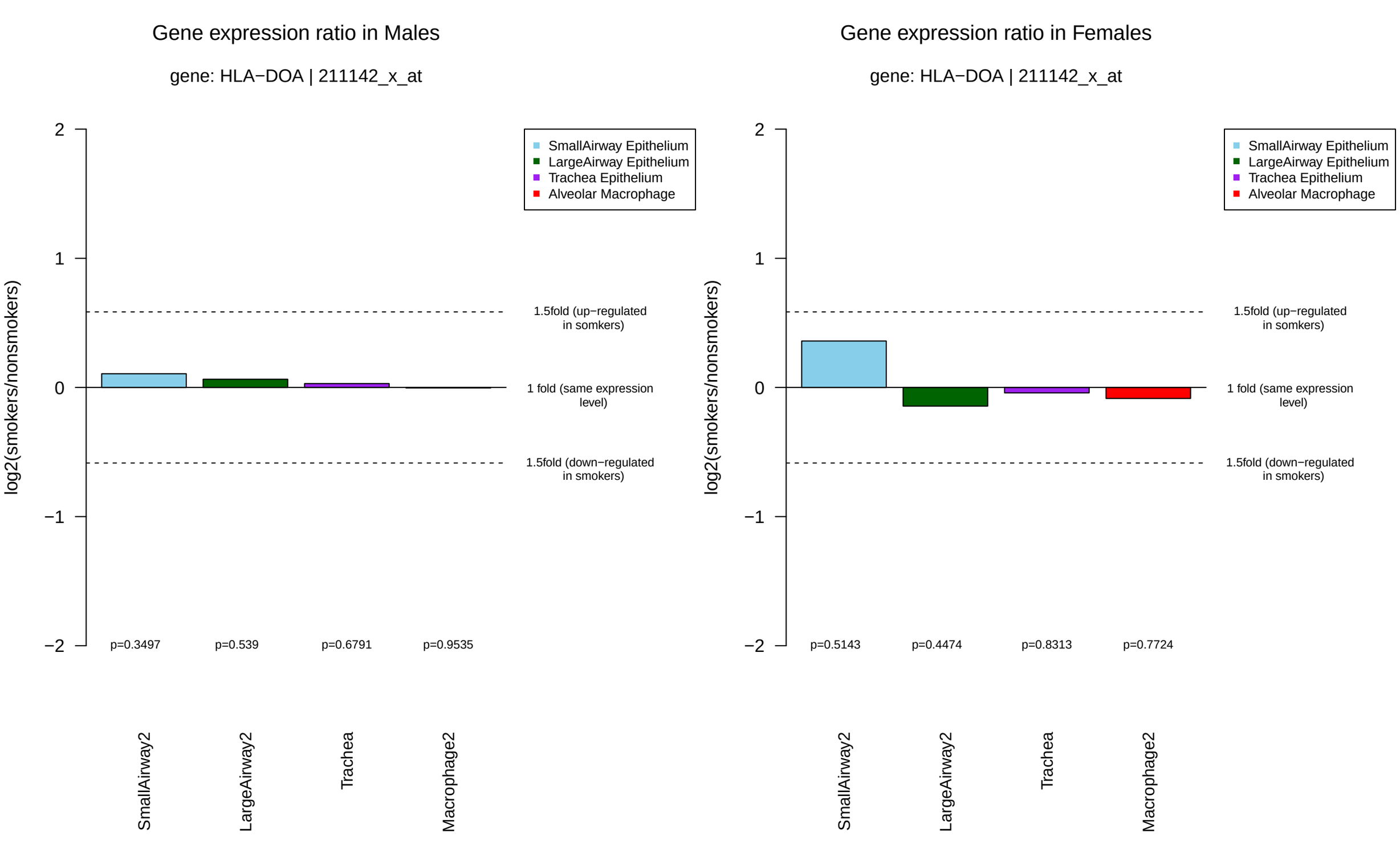
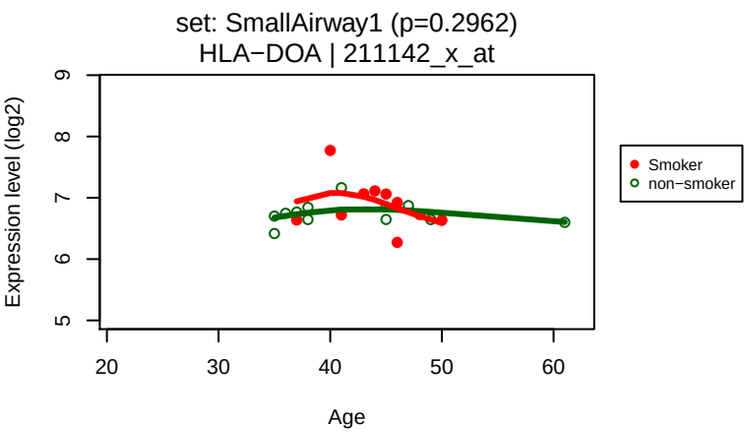
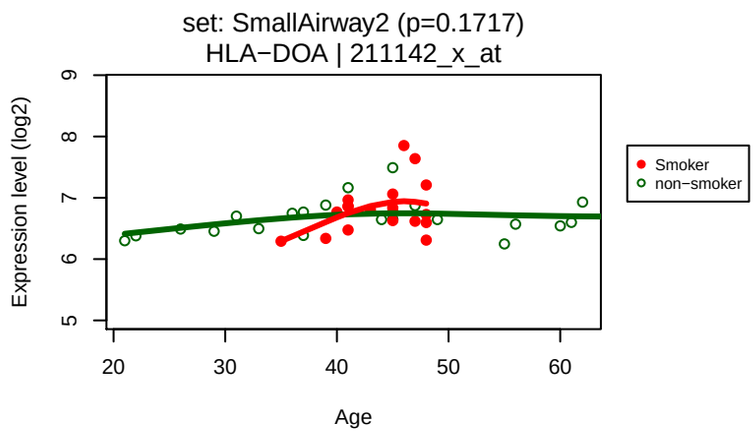
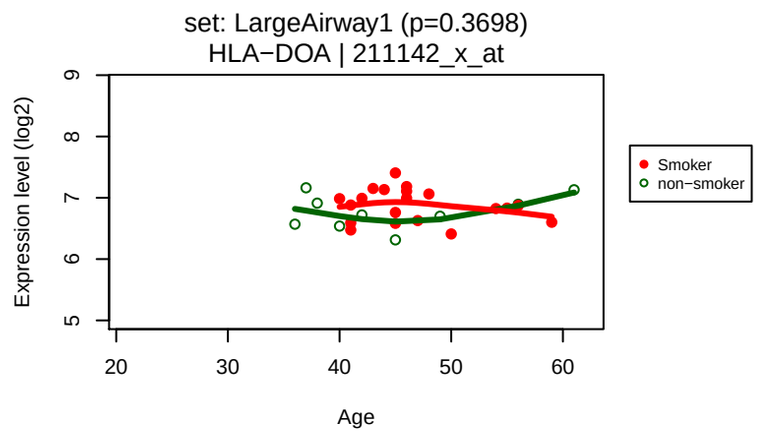

Gene expression of smokers and nonsmokers
gene: HLA−DOA | 211142_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2962
p=0.1717
p=0.2166
p=0.3698
p=0.92
p=0.8886
p=0.4713
p=0.7151
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

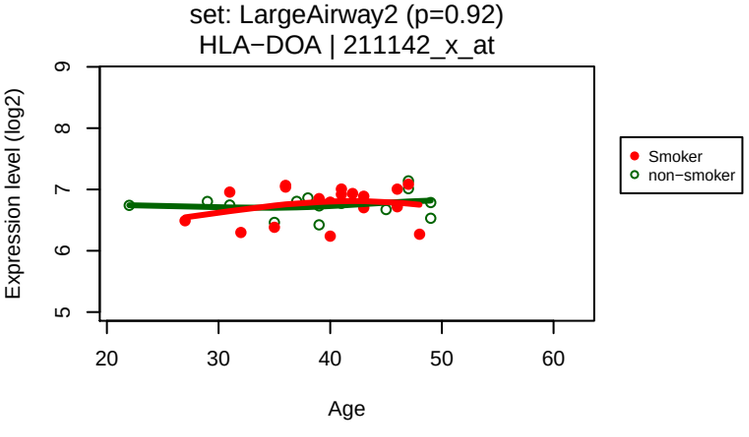

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DOA | 211142_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2962
p=0.1717
p=0.2166
p=0.3698
p=0.92
p=0.8886
p=0.4713
p=0.7151
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

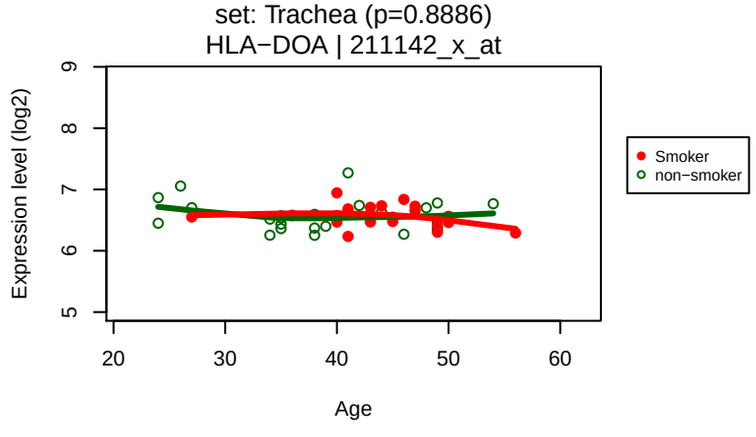

Gene expression ratio in Males
gene: HLA−DOA | 211142_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3497
p=0.539
p=0.6791
p=0.9535
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DOA | 211142_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5143
p=0.4474
p=0.8313
p=0.7724
SmallAirway2
LargeAirway2
Trachea
Macrophage2

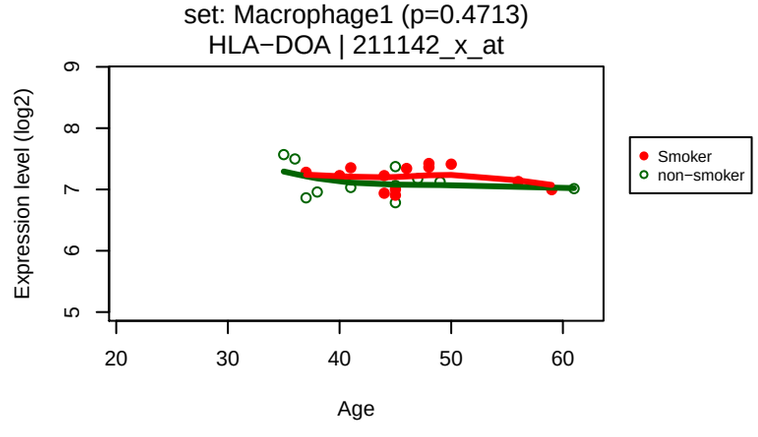

set: SmallAirway1 (p=0.2962)
HLA−DOA | 211142_x_at
Expression level (log2)
Age
Smoker
non−smoker

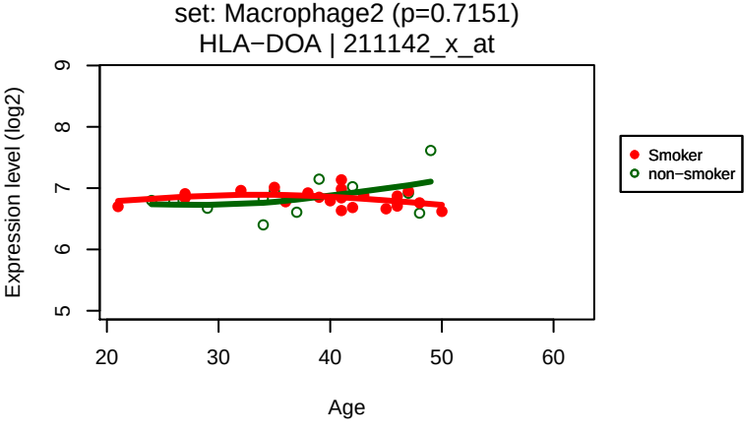

set: SmallAirway2 (p=0.1717)
HLA−DOA | 211142_x_at
Expression level (log2)
Age
Smoker
non−smoker

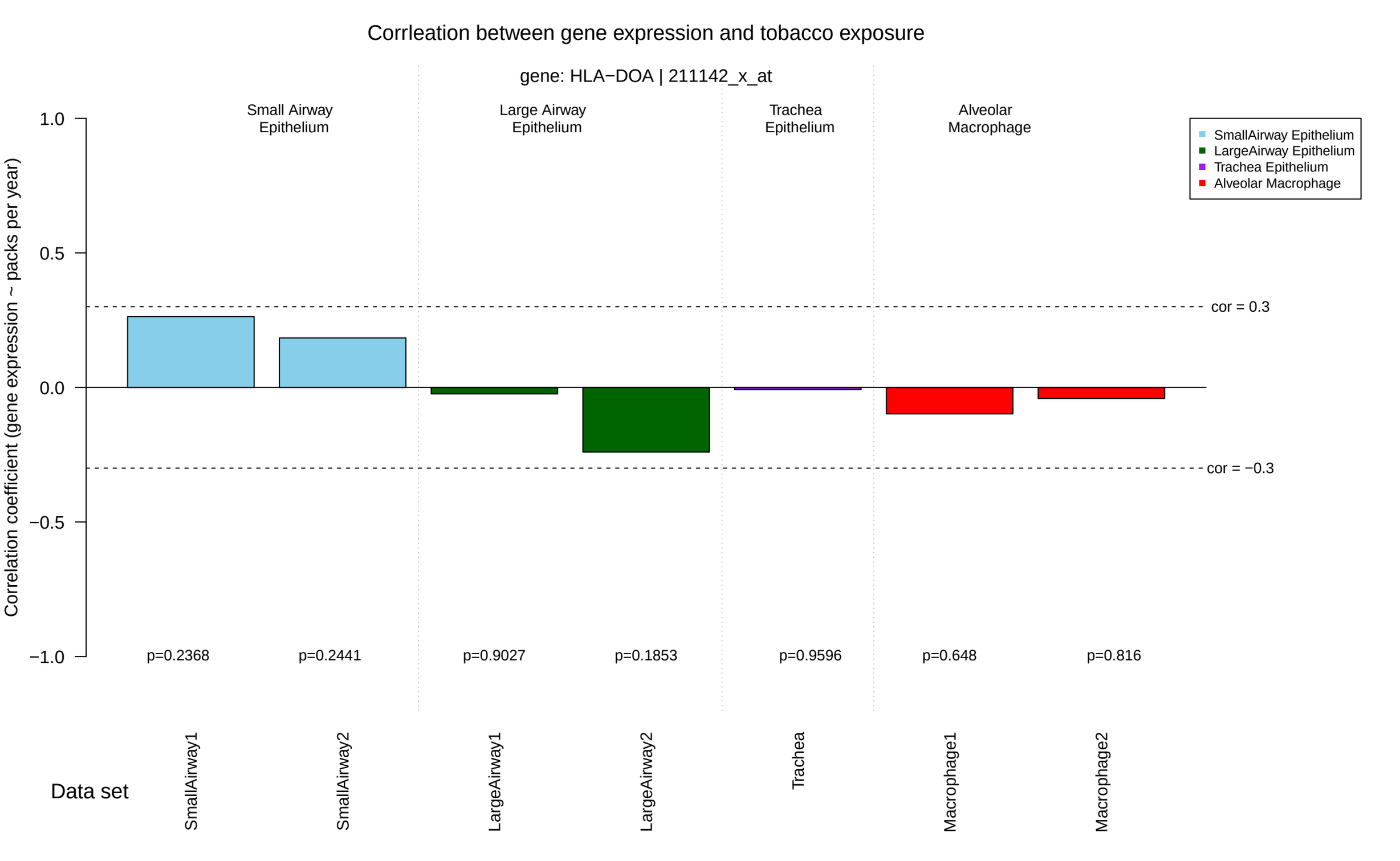

set: LargeAirway1 (p=0.3698)
HLA−DOA | 211142_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.92)
HLA−DOA | 211142_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8886)
HLA−DOA | 211142_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4713)
HLA−DOA | 211142_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7151)
HLA−DOA | 211142_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DOA | 211142_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2368
p=0.2441
p=0.9027
p=0.1853
p=0.9596
p=0.648
p=0.816
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2